গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহ্জালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING
AT MIAH FAZILCHIST, SUBIDBAZAR, SYLHET

DAG # SA-501, BS-2538, KHA # SA-2019, MUTATION-18282
J.L # SA-91, BS-76, MOUJA- MUNICIPALITY

Verification Code: dw2QVCd1
Verify report at: crtc-cee-sust.com

CLIENT

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CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
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SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

THE SOIL ENGINEERS & CONSULTANT
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-55 ft	11
02	Disturbed sample	02	0-60 ft	12
03	Disturbed sample	03	0-60 ft	12
04	Disturbed sample	04	0-60 ft	12
05	Disturbed sample	05	0-60 ft	12
06	Disturbed sample	06	-	-
07	Disturbed sample	07	-	-
08	Disturbed sample	08	-	-
09	Disturbed sample	09	-	-
10	Disturbed sample	10	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	6	0.696	0.732	0.798	0.822	0.888	0.966	0.750
	5.0-10	9	1.084	1.140	1.243	1.280	1.383	1.505	1.168
BH02	2.5-5.0	6	0.650	0.683	0.745	0.767	0.829	0.902	0.700
	5.0-10	7	0.852	0.896	0.977	1.006	1.087	1.183	0.918
BH03	2.5-5.0	10	1.160	1.220	1.330	1.370	1.480	1.610	1.250
	5.0-10	10	1.128	1.186	1.293	1.332	1.439	1.566	1.216
BH04	2.5-5.0	5	0.626	0.659	0.718	0.740	0.799	0.869	0.675
	5.0-10	8	0.936	0.984	1.073	1.105	1.194	1.299	1.009
BH05	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	5	0.536	0.564	0.615	0.633	0.684	0.744	0.578
BH06	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH07	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH08	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH09	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BH10	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.096	0.138	0.096	0.074	0.032					
10	0.196	0.156	0.174	0.138	0.048					
15	0.096	0.096	0.246	0.138	0.096					
20	0.120	0.239	0.048	0.096	0.096					
25	0.118	0.096	0.118	0.120	0.174					
30	0.174	0.218	0.174	0.096	0.118					
35	0.118	0.218	0.118	0.074	0.138					
40	0.156	0.261	0.156	0.138	0.138					
45	0.218	0.196	0.048	0.156	0.156					
50	0.200	0.239	0.136	0.136	0.200					
55	0.192	0.192	0.192	0.192	0.192					
60		0.184	0.184	0.184	0.184					
65										
70										
75										
80										
85										
90										
95										
100										

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.900	1.350	0.900	0.675	6.400					
10	2.025	1.575	1.800	1.350	9.600					
15	0.900	0.900	3.375	1.350	0.900					
20	24.000	2.475	9.600	19.200	0.900					
25	1.125	0.900	1.125	24.000	1.800					
30	1.800	2.250	1.800	0.900	1.125					
35	1.125	2.250	1.125	0.675	1.350					
40	1.575	2.700	1.575	1.350	1.350					
45	2.250	2.025	9.600	1.575	1.575					
50	40.000	2.475	27.200	27.200	40.000					
55	38.400	38.400	38.400	38.400	38.400					
60	-	36.800	36.800	36.800	36.800					
65										
70										
75										
80										
85										
90										
95										
100										

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.062	0.090	0.062	0.048	0.021					
10	0.127	0.101	0.113	0.090	0.031					
15	0.062	0.062	0.160	0.090	0.062					
20	0.078	0.156	0.031	0.062	0.062					
25	0.077	0.062	0.077	0.078	0.113					
30	0.113	0.141	0.113	0.062	0.077					
35	0.077	0.141	0.077	0.048	0.090					
40	0.101	0.170	0.101	0.090	0.090					
45	0.141	0.127	0.031	0.101	0.101					
50	0.130	0.156	0.088	0.088	0.130					
55	0.125	0.125	0.125	0.125	0.125					
60		0.120	0.120	0.120	0.120					
65										
70										
75										
80										
85										
90										
95										
100										

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.405	0.608	0.405	0.304	2.880					
10	0.911	0.709	0.810	0.608	4.320					
15	0.405	0.405	1.519	0.608	0.405					
20	10.800	1.114	4.320	8.640	0.405					
25	0.506	0.405	0.506	10.800	0.810					
30	0.810	1.013	0.810	0.405	0.506					
35	0.506	1.013	0.506	0.304	0.608					
40	0.709	1.215	0.709	0.608	0.608					
45	1.013	0.911	4.320	0.709	0.709					
50	18.000	1.114	12.240	12.240	18.000					
55	17.280	17.280	17.280	17.280	17.280					
60		16.560	16.560	16.560	16.560					
65										
70										
75										
80										
85										
90										
95										
100										

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed six-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is quite irregular in between the boreholes and also remarkable vertical inconsistency exists in the soil formation.

In most boreholes, gray soft clay with low bearing capacity exists roughly within 15-45 ft from EGL followed by a fine sand layer of about 10 ft (as the client provided) with high bearing capacity.

Foundation options

In this situation, the whole site may be densified by sand/wick drain method. The number, size, and depth of the sand/wick drain should be selected by the foundation/design engineer considering both the pressure influence zone and the degree of densification required to minimize uneven settlement. After soil improvement, the appropriate pattern of the shallow foundation either in form of individual or continuous, or even raft may be selected on the basis of the degree of ground improvement. Although bearing capacity values for designing a shallow foundation can be found in the attached Table (7) in the report which is based on existing soil conditions, it is necessary to reinvestigate the modified bearing capacity of the densified stratum after ground improvement.

Safety provisions

Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **55-60** ft as the client provided.

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